Semester Project Milestone 5: Rapid Prototyping & Draft Testing Plan

Project Title: SpoilGuard
Group ID: 12
Team Members: Isaac, Kofi, Nti, Igdel
Prototype Link: https://github.com/ntious/FundIT/blob/main/final_project/spoilguard.html

1. Project Overview

SpoilGuard is a mobile application designed to reduce household food waste by helping users track groceries, monitor expiration dates, and respond to timely alerts. Many families over-purchase and lack visibility into what they already have, leading to avoidable waste with both financial and environmental consequences. Existing solutions often fall short due to heavy reliance on manual entry and limited support for in-home food management. SpoilGuard addresses this gap by combining OCR-based receipt scanning, inventory tracking, expiration reminders, recipe suggestions, and simple analytics. The primary users are busy adults managing household groceries, with additional consideration for accessibility needs such as older adults. In this milestone, we developed a visual prototype demonstrating the four core use cases from Milestone 4 and prepared a draft testing plan for the next phase.

2. User Journey Demonstration (Written Walkthrough)

This written walkthrough demonstrates how users complete the four major use cases through the prototype. The prototype includes an app dashboard, a receipt-scanning workflow, fridge inventory management, an alerts view, analytics reporting, recipes, and accessibility settings. The HTML prototype used for this sample supports interactive screen changes and simulated app behavior.

Use Case 1: Automated Inventory Logging via Receipt Scan

Sarah, a busy parent returning from a grocery trip, opens the app on the Dashboard screen (Figure 1). She selects Scan Receipt, which opens the Scan Receipt screen showing a camera placeholder and capture instructions (Figure 2). After selecting Capture Receipt, the app simulates scanning and transitions to the OCR Results screen, where detected grocery items such as eggs, milk, strawberries, and spinach are displayed for confirmation (Figure 3). Sarah keeps the detected items selected and chooses Add to My Fridge. The system then updates the inventory and takes her to the My Fridge screen, where the newly added items are visible.

Use Case 2: Manual Inventory Update

From the My Fridge screen (Figure 4), Sarah can review all active grocery items and see how long each item has before expiration. Each inventory card includes action buttons such as Consumed and Discarded. For example, if she uses spinach in dinner, she can select Consumed, and the system removes it from the active inventory. If an item has already spoiled, she can select Discarded, and the system updates both the inventory and waste analytics.

Use Case 3: Smart Expiration Alerts

When items approach expiration, users can access the Alerts screen from the dashboard or bottom navigation (Figure 5). The system groups urgent items first, such as eggs or milk with one to two days remaining. Each alert card shows the item name, urgency level, and a recommended action such as Use today or Use soon. When the user selects an item, the app reveals more detailed information about the item, including its remaining shelf life and suggested response.

Use Case 4: View Savings and Waste Analytics

Users can open the Analytics screen from the dashboard or bottom navigation (Figure 6). This screen summarizes total savings, items consumed, items discarded, and current tracked items. It also displays a simple waste breakdown using progress bars and a visual monthly trend. This screen helps users understand their grocery behavior in a more meaningful way. Instead of just seeing alerts, they can see whether their behavior is improving over time.

Accessibility and Settings

The Settings screen (Figure 7) allows users to enable high contrast mode, large font mode, and audio alerts. These settings directly reflect the refinements made in Milestone 4, especially the needs identified through the Robert edge-case scenario. This screen ensures the app is not only functional but also more inclusive.

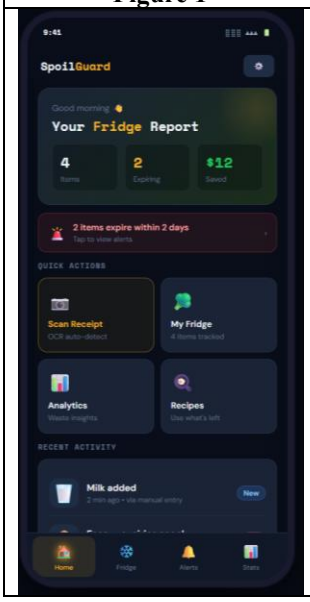
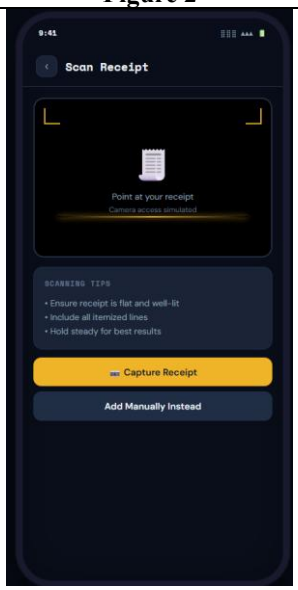
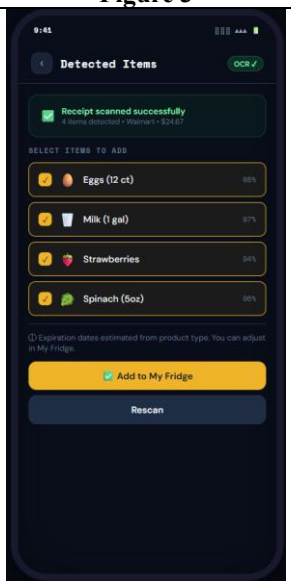
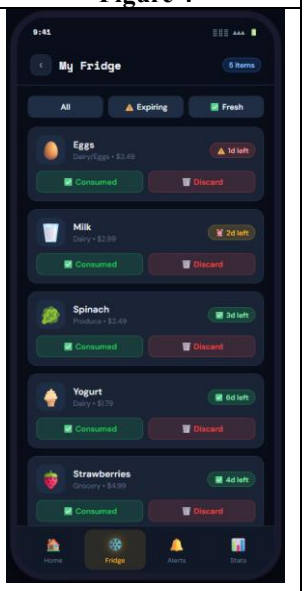
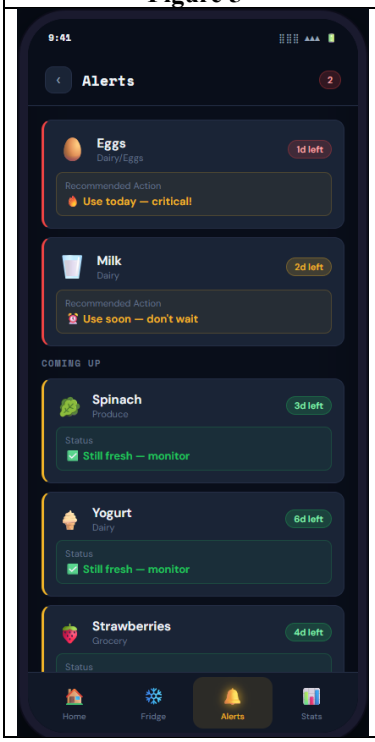
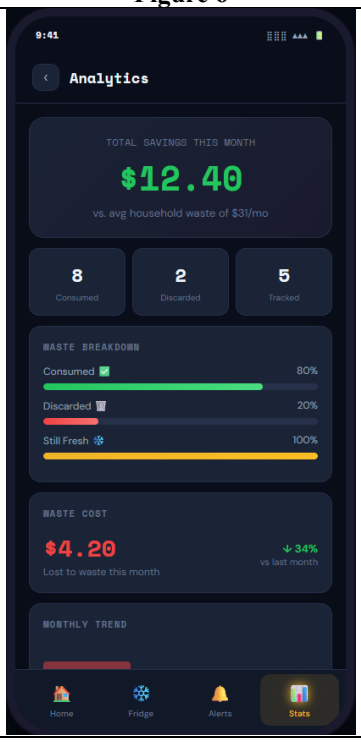
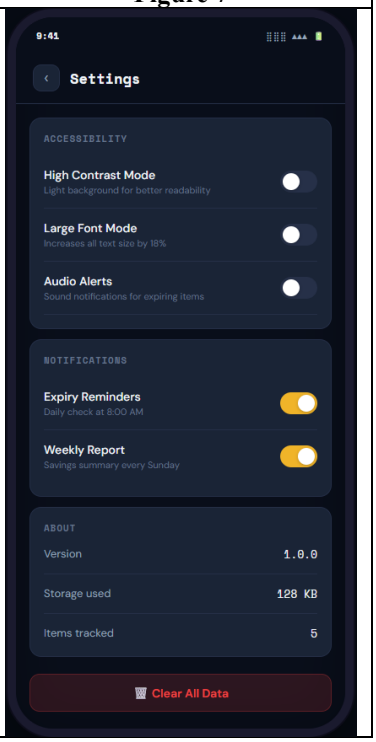
Figure 1	Figure 2	Figure 3	Figure 4
			

Figure 5	Figure 6	Figure 7
		

3. Prototype Submission

Prototype Format

This sample prototype was created as a single-file standalone HTML interface that can be opened by double-clicking the file in a browser. It simulates a mobile application and includes:

- Home / Dashboard
- Scan Receipt
- OCR Results
- My Fridge
- Alerts
- Analytics
- Recipes
- Settings

The uploaded prototype file shows that the interface includes simulated navigation, inventory state updates, OCR item confirmation, inventory removal actions, alert panels, analytics summaries, and accessibility toggles.

Prototype Access

Prototype File: spoilguard.html

Prototype Type: Zero-setup interactive HTML prototype

How to Run: Open the file in any modern web browser by double-clicking it.

Prototype Completeness Check

All screens referenced in the walkthrough are included in the prototype:

- Figure 1: Dashboard
- Figure 2: Scan Receipt
- Figure 3: OCR Results
- Figure 4: My Fridge
- Figure 5: Alerts
- Figure 6: Analytics
- Figure 7: Settings

4. Draft Testing Plan

The goal of testing in Milestone 6 will be to observe whether real users can complete the major SpoilGuard flows with minimal confusion and whether the interface supports both usability and accessibility.

Testing Component	**Draft Plan**
Location	Testing will take place in the Engineering Library study area and, if needed, through a synchronous Zoom session for remote testers.
Participants	We plan to recruit 5–6 participants. Our preferred users are college students and young adults who regularly shop for groceries, since they are similar to one of our target user groups. We would also like at least 1 participant who benefits from larger text or accessibility support.
Introduction Script	"Thank you for helping us test our prototype. This is a food waste reduction app called SpoilGuard. Please interact with it as naturally as possible and think aloud if you get confused or unsure about what to do next."
Task Scenario	"You just came home from the grocery store. Your goal is to scan your receipt, add detected items to your fridge, and then check whether anything is close to expiring."
Metric / Method	We will measure time to completion, number of incorrect clicks, whether the participant completes the task without help, and major confusion points noted during observation.

Expected Insight	We want to learn whether users can understand the receipt-scanning flow easily, whether the dashboard makes the next step obvious, and whether alert information is easy to interpret.
Rationale	These methods are appropriate because Milestone 5 focuses on prototype usability rather than technical implementation. Task completion and confusion points will help us identify interface problems before more advanced development or testing.

Additional Planned Testing Tasks

To align with all four use cases, we may also test:

1. Marking a fridge item as consumed
2. Identifying which item is most urgent from the alerts screen
3. Interpreting the analytics summary
4. Enabling accessibility settings such as large font mode

These extra tasks will help us determine whether the system supports a full range of user interaction rather than only the scanning workflow.

5. Group Participation

- **Isaac:** Helped translate the Milestone 4 use cases into prototype flows and reviewed whether the dashboard and alert features reflected the original problem and user needs.
- **Kofi:** Led the structure of the milestone submission, connected the prototype to the four use cases, and drafted the written walkthrough and testing plan.
- **Nti:** Reviewed prototype screen logic and helped verify that the inventory, alerts, and analytics screens aligned with the functional requirements.
- **Igdel:** Reviewed the prototype presentation format, checked the completeness of the figures/screens, and edited the final document for clarity and organization.

All group members contributed to milestone completion. No contract violations occurred during this phase.

6. AI Disclosure

Our group used Claude AI tools in a limited, disclosed manner to support prototype planning and drafting. AI was used to help:

- turn Milestone 4 use cases into screen flow ideas,
- suggest prototype structure for a zero-setup HTML version,
- and improve wording in the draft walkthrough and testing plan.

However, the project topic, requirements, use cases, testing rationale, and final submission structure were determined and reviewed by the group. The final prototype and report were refined to match the SpoilGuard project scope and Milestones 2–4. The prompt used is in Appendix A.

Appendix A: Final Prompt for Generating SpoilGuard HTML Prototype (Zero-Setup)

You are an expert front-end developer and UX designer.
Your task is to generate a COMPLETE, SINGLE-FILE HTML prototype that simulates a mobile application called "SpoilGuard".
The output MUST:
- Be ONE standalone HTML file
- Use only HTML, CSS, and minimal JavaScript
- Run by double-clicking (no installation, no dependencies)
- Simulate navigation between screens
- Be clean, readable, and well-structured
- NOT require React, frameworks, or external libraries

PROJECT CONTEXT

App Name: SpoilGuard
Problem: Households waste 30–40% of food due to poor tracking of groceries and expiration dates.
Solution:
A mobile app that:
- Scans grocery receipts (OCR simulation)
- Tracks expiration dates
- Sends alerts for expiring items
- Provides analytics on food waste and savings

Target Users:
- Busy parents (fast interaction needed)
- Elderly users (accessibility support needed)

FUNCTIONAL REQUIREMENTS

FR-01: Manual grocery logging
FR-02: Receipt scanning (OCR simulation)
FR-03: Expiration tracking and alerts
FR-04: Recipe suggestions
FR-05: Inventory updates (consume/discard)
FR-06: Analytics dashboard

USE CASES (MUST ALL BE IMPLEMENTED)

UC-01: Scan Receipt (OCR Simulation)
Flow:
- User clicks "Scan Receipt"
- Camera placeholder appears
- User clicks "Capture"
- System shows detected items
- User confirms → items added to inventory
UC-02: Manage Inventory (My Fridge)
Flow:
- User views inventory list
- Each item shows expiration
- User can mark item as:
  - Consumed

  - Discarded
UC-03: Expiration Alerts
Flow:
- System highlights items expiring soon
- User clicks alert → sees item details
- Show recommended action ("Use soon")
UC-04: Savings Analytics
Flow:
- User views summary:
  - Items consumed
  - Items discarded
  - Waste cost
- Show simple visual (progress bar)

SCREEN REQUIREMENTS

You MUST include these screens:
1. Dashboard (Home / Hub)
2. Scan Receipt Screen
3. OCR Results Screen
4. My Fridge Screen
5. Alerts Screen
6. Analytics Screen
7. Settings Screen (Accessibility)

UI DESIGN REQUIREMENTS

Style:
- Mobile-app layout centered on screen
- Clean, modern UI
- Card-based design

Colors (High Contrast Mode):
- Background: dark (#0f172a)
- Card: #1f2937
- Accent: yellow (#facc15)
- Text: white/light
Components to include:
- Buttons (primary + secondary)
- Cards
- Inventory list items
- Badges (expiration warning)
- Toggle switches (settings)

INTERACTIVITY REQUIREMENTS

You MUST implement basic JavaScript to simulate:
Navigation:
- Clicking buttons switches screens
- Include back button
State:

- Maintain a simple inventory array
- When "Confirm Items" is clicked → items appear in "My Fridge"
Actions:
- "Consumed" → remove item
- "Discarded" → remove item and increase waste cost

Alerts:
- Highlight items expiring in ≤ 2 days
Settings:
- Toggle:
  - High contrast mode (light/dark)
  - Large font mode
  - Audio alerts (visual indicator only)

DATA (USE THIS INITIAL DATA)

Inventory:
- Eggs (expires in 1 day)
- Milk (expires in 2 days)
- Spinach (expires in 3 days)
- Yogurt (expires in 6 days)

OCR Detected Items:
- Eggs
- Milk
- Strawberries
- Spinach

ACCESSIBILITY REQUIREMENTS

- Large readable text
- High contrast UI
- Clearly labeled buttons
- Simple navigation

OUTPUT FORMAT (STRICT)

Return ONLY the HTML code.
DO NOT:
- Add explanations
- Add comments outside HTML
- Use external libraries
- Use multiple files
The result must be:
- Fully functional when opened directly in a browser
- Visually structured like a mobile app
- Easy to understand for demonstration
GOAL: Produce a professional, classroom-ready interactive prototype that demonstrates all 4 use cases clearly.